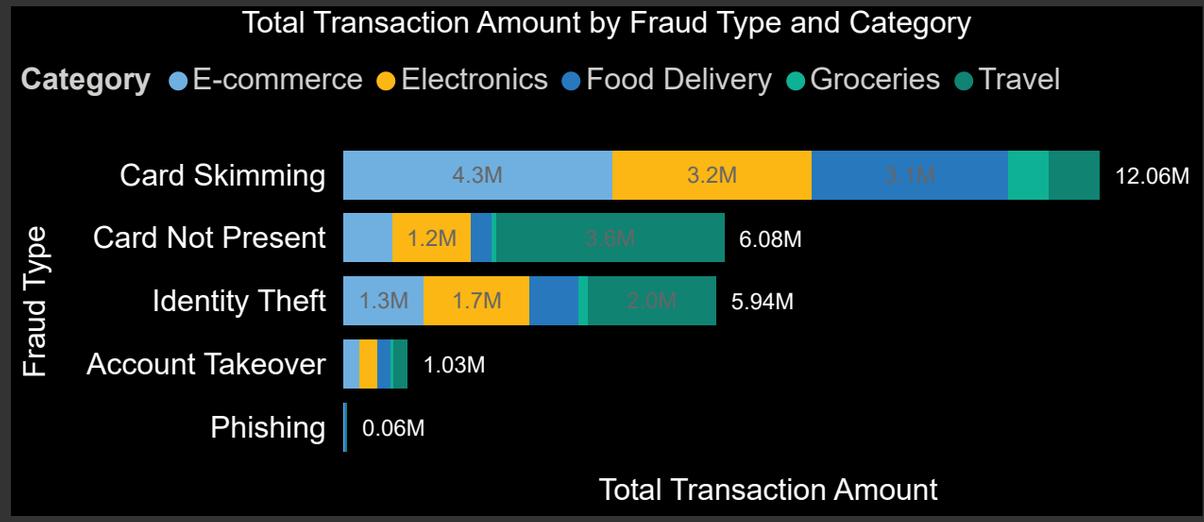
Credit Card Fraud Risk Analysis

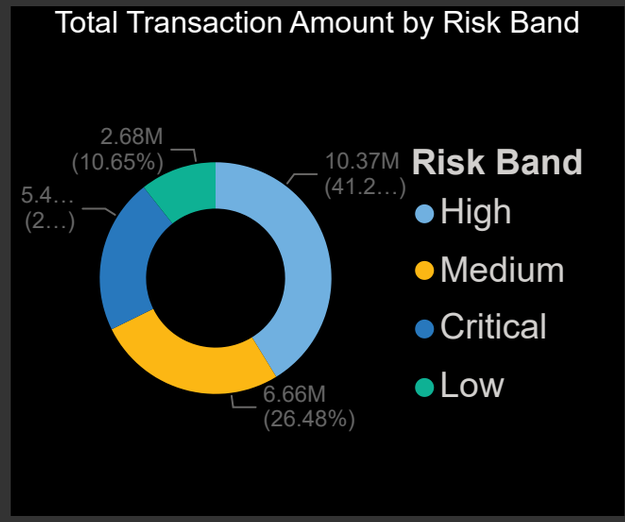
Fraud Type
All

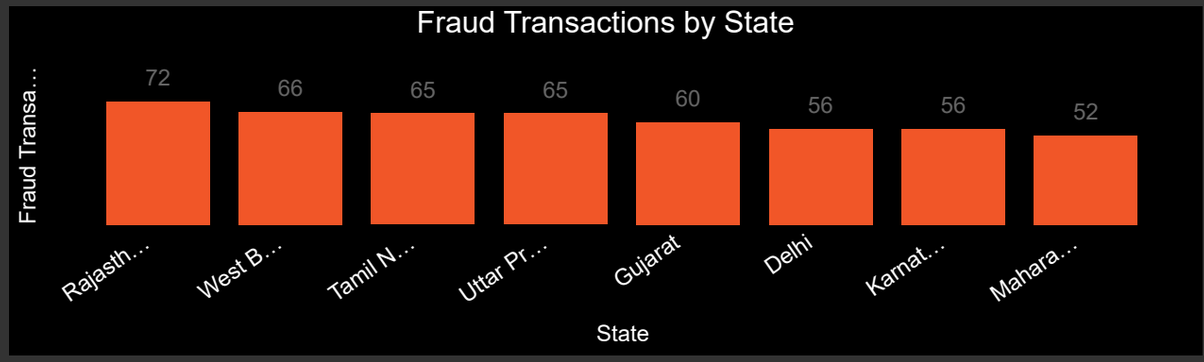
State
All

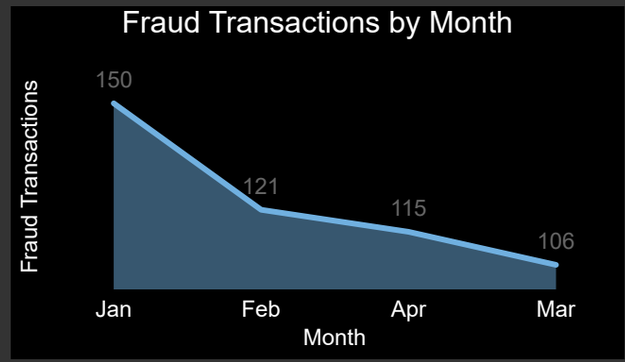
Merchant
All

285K	492	25.16M	60.13K
Total Transactions	Fraud Transactions	Total Trnx Amount	Total Fraud Amount









Credit Card Fraud Risk Analysis

Fraud Type
All

State
All

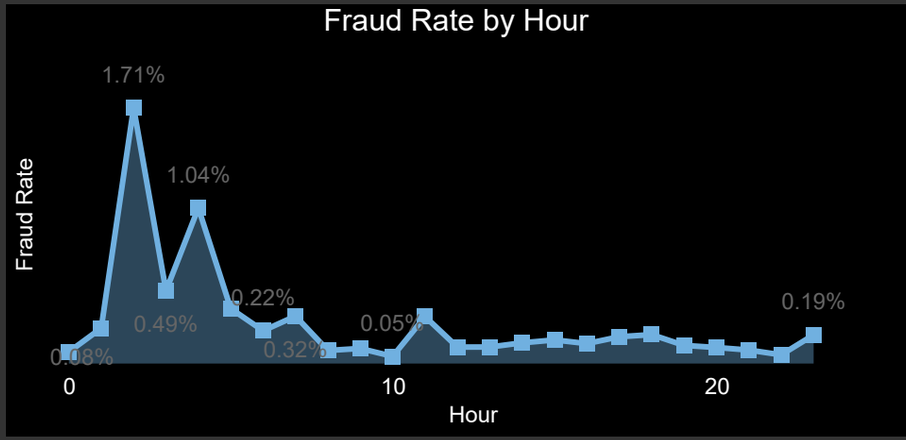
Merchant
All

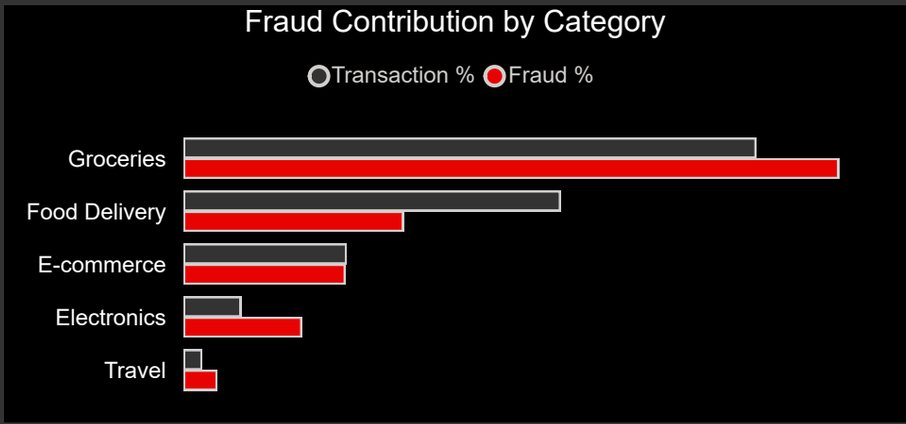
0.17%
Fraud Rate

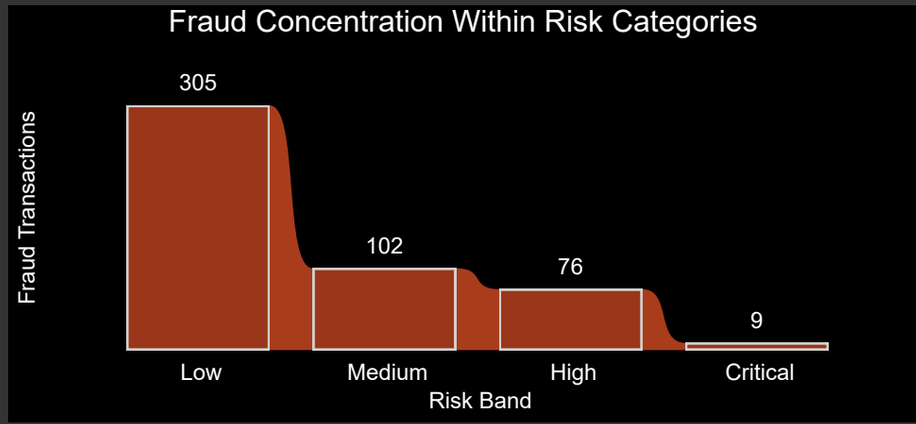
2 AM
Peak Fraud Time

15.45%
High Risk Contribution

$122
Avg Fraud Amount







Credit Card Fraud Risk Analysis

Category
All

Hour
All

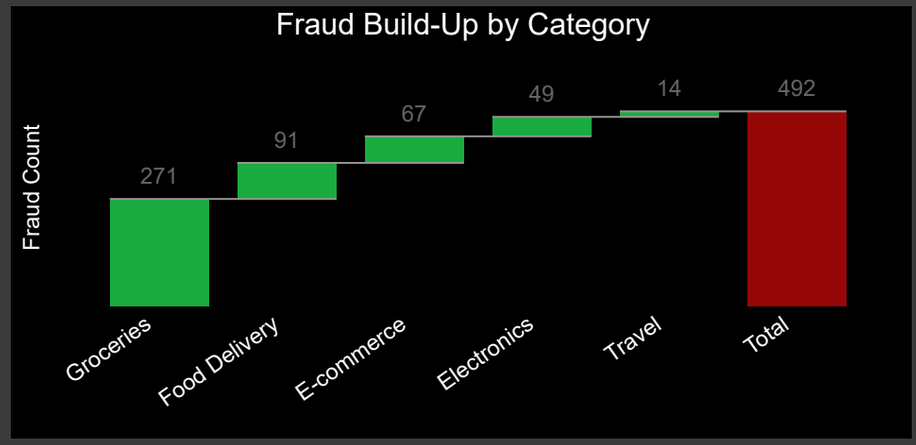
Transaction Type
All

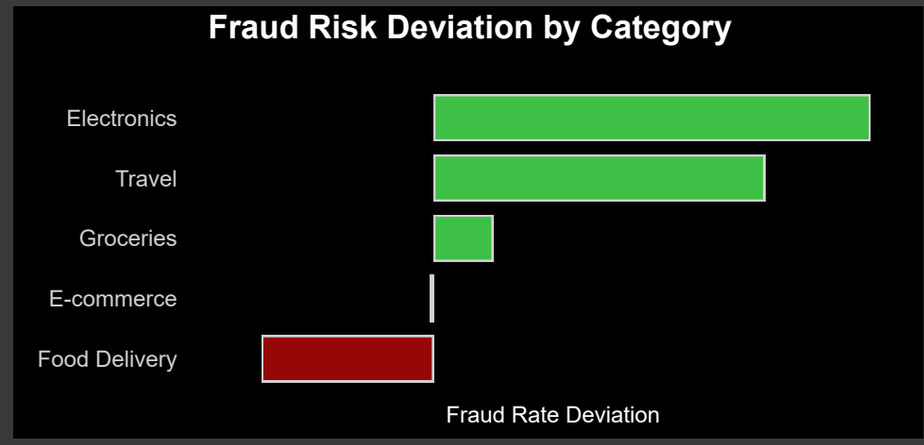
55.08%
Fraud Concentr. Index

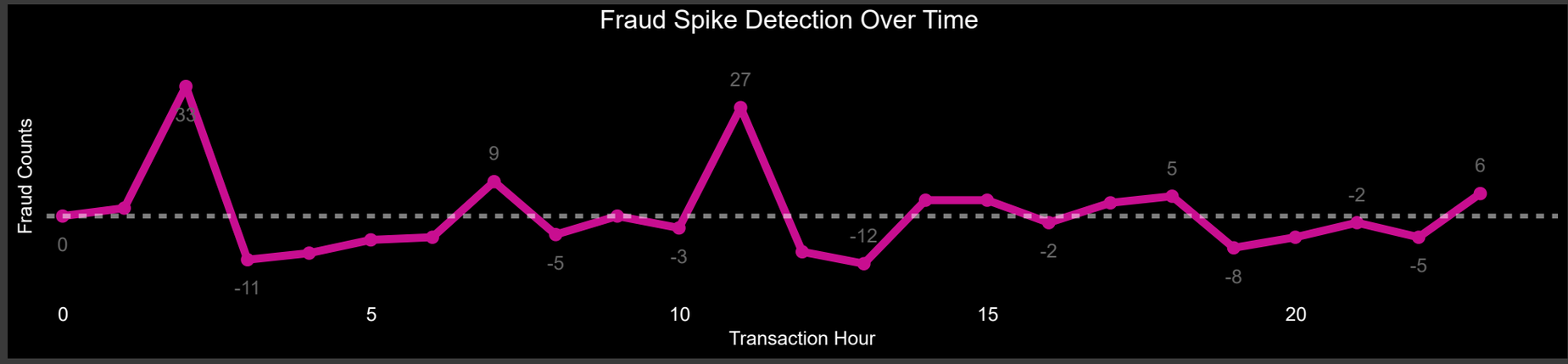
11.59%
Time Skew Index

1.38
Fraud Intensity Ratio

74%
Repeat Fraud Ratio







Fraud Insights & Strategic Actions

High-Loss Fraud Types

What: Few fraud types drive most losses
Why: Weak authentication in high-value channels
Action: Risk-based verification for high-value transactions

Geographic Risk Concentration

What: Fraud clustered in select states
Why: Uneven controls & merchant quality
Action: Region-specific limits & monitoring

Risk Band Optimization

What: High/Critical = majority of fraud
Why: Risk scores not used for decisions
Action:
- High → Verify/Block
- Medium → Monitor

Time-Based Fraud Peaks

What: Fraud spikes at specific hours
Why: Low monitoring during off-hours
Action: Dynamic controls by time window

Category & Merchant Risk

What: Specific categories & merchants dominate fraud
Why: High-value targets + weak oversight
Action: Category-level checks & Flag High-Risk merchants.

Fraud Growth Momentum

What: Fraud increases in bursts, hints scaling behavior rather than isolated events
Why: Once a method succeeds, it is rapidly repeated across similar transactions
Action:
- Track **rate of change in fraud (MoM / hourly spikes)**
- Implement **early intervention triggers**

📌 Strategic Conclusion & Recommendation

Fraud patterns are **predictable and concentrated across type, geography, time, and risk segments**.
By shifting from uniform controls to **targeted, risk-based interventions**, organizations can significantly reduce fraud losses while maintaining operational efficiency and customer experience.